

Self-Supervised Deep Clustering Method for Detecting Abnormal Data of Wastewater Treatment Process

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Abstract—In wastewater treatment process (WWTP), abnormal data seriously reduce data quality rendering the application techniques impractical. The implementation of abnormal data detection is challenging due to the nonlinear nature of WWTP. Typically, constructing an accurate anomaly detector requires large amounts of labeled data, which is difficult in practice. Thus, a self-supervised memory enhanced deep clustering method (SMEL) is proposed to detect abnormal data without using any labels. First, a self-supervised deep clustering network, combining stacked autoencoders and the clustering algorithm, leverages unlabeled data to extract nonlinear features and capture the normal pattern. Second, an adaptive weight objective function, jointly optimizing the reconstruct error and clustering error, is designed to obtain a robust clustering structure. Third, double memory enhanced modules, consisting of a centroid memory and a score memory, are presented to enhance training stability and detection accuracy. Finally, experiments on three WWTP datasets show that SMEL achieves the highest detection accuracy.

Index Terms—Abnormal data detection, adaptive weight objective function, double memory enhanced modules, self-supervised deep clustering network, wastewater treatment process (WWTP).

Manuscript received 26 December 2022; revised 11 March 2023; accepted 8 April 2023. Date of publication 20 April 2023; date of current version 19 January 2024. This work was supported by National Key Research and Development Project under Grants 2022YFB3305800-5, National Science Foundation of China under Grants 62125301, 61890930-5, 61903010, and 62021003, Beijing Outstanding Young Scientist Program under Grant BJJWZYJH01201910005020, and in part by the Beijing Natural Science Foundation under Grant KZ202110005009, and Youth Beijing Scholar under Grant 037. (Corresponding author: Honggui Han.) Paper no. TII-22-5195.

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2023.3268777>.

Digital Object Identifier 10.1109/TII.2023.3268777

I. INTRODUCTION

WASTEWATER treatment process (WWTP) plays an essential role in removing the organic wastes and producing the renewable water source [1]. With recent progress in advanced techniques, WWTP has been equipped with various sensors and transmission equipment for collecting, transmitting, and storing time series. These time series, implying the operation status of WWTP, have strong nonlinearity and time-varying characteristic. To improve effluent quality and guarantee stable operation, it is necessary to analyze the data generated by the treatment process and obtain the relevant information about its operation status [2].

Abnormal data of WWTP, which are caused by various reasons, have been an inevitable problem [3]. Specifically, abnormal data are caused by working condition faults or transmission faults, among which working condition faults have been widely studied. Abnormal data in WWTP caused by transmission faults are rarely mentioned. The article mainly focuses on the transmission faults, such as the disturbance of unstable current, sensor faults, communication losses, and changed environment. Abnormal data are significantly different from or inconsistent with the normal existing data. The existence of abnormal data makes the distribution of training data deviate, which results in the model unable to extract normal features [4]. Moreover, as WWTP demands stability and reliability, maintainers need to detect abnormal operations and implement specific maintenance operations [5]. Thus, to detect abnormal data, one straightforward idea is that abnormal data detection method is used.

In the past few decades, the existing abnormal data detection studies can be broadly divided into three categories: statistical-based, model-based, and clustering-based approaches. The statistical-based method assumes that the data follow a predetermined distribution or probability model and identifies abnormal values that deviate from such distribution or are in the region of low probability [6], [7], [8]. These statistical-based methods always assume the normal data follow a certain of data distribution. Nevertheless, we cannot always have this kind of prior data distribution knowledge in practice, especially for high dimensional real data sets. For clustering-based methods, these methods calculate the corresponding anomaly scores based on the density information of the data point and its neighboring data and finally compare the value with a preset threshold to determine whether the data are abnormal [9], [10], [11]. The

clustering-based methods have two main limitations. First, these methods assume that the shape and form of the clusters are considered. Second, as the number of dimensions of the data increases, the detection performances of these methods decrease. As a promising method, the model-based methods, which use a predictive model to characterize the normal data and then detect abnormal values as deviations from the model, have been studied in recent years [12], [13]. Shakiba et al. [14] addressed fault diagnosis for transmission lines using the convolutional neural network. The data used in [14] were generated due to device faults. However, the abnormal data studied in this article are caused by transmission faults due to the unstable working environment. Although the above model-based methods have achieved good results in abnormal data detection, these methods rely too much on manually setting of attribute relationship. Human intervention is more serious, limiting its robustness and generalization ability. In addition, these methods are shallowly structured, limiting their learning and classification capabilities. Since WWTP data are complex with nonlinearity and time-varying characteristic, these methods may not be flexible and accurate enough.

As a part of model-based methods, deep learning methods with the strong ability in nonlinear extraction and data modeling have obtained great achievements in the field of abnormal data detection. Patnaik et al. [15] combined the convolutional network and the long short-term memory network to construct a supervised data extraction system for detecting automated web page. Pan et al. [16] proposed a multiscale analysis method for rolling bearing fault diagnosis by integrating the inner product for capturing feature and the kernels with different sizes for generating better performances. Yu et al. [17] presented a novel deep common spatial pattern model to transform data into another mapping with data of different categories. These categories have maximal differences in their measures of dispersion. However, the performance of feature extraction was not satisfactory. Luo et al. [18] adopted stacked recurrent neural networks to detect anomalies by iteratively updating the spare coefficients. However, it demanded a robust mechanism to generate large reconstruction error on the anomalies. Wang et al. [19] presented a stacked autoencoders (SAE) model for fault classification to address the problem of deep features contain information unrelated to faults. Hu et al. [20] utilized a stacked sparse autoencoders model to learn generic driving behavior features and introduced the dropout technology to avoid overfitting. In [21], a convolutional deep neural network was presented for one-class classification. In the method, the labeled data from an unrelated task were utilized to perform feature learning. Though much progress has been done for abnormal data detection, most of the existing works always assume that training data are labeled for building the abnormal data detection model or classifier. However, it is difficult and time-consuming to obtain a great quantity of labeled data required in the training process.

To detect abnormal data without using any labels, the integration of clustering-based and deep learning method has been considered as a promising solution. Lu and Hanjalic [22] presented a context-dependent scaling factors based spectral clustering method that was utilized to detect acoustic events

from audio streams. Lin et al. [23] proposed a density clustering algorithm employed on pretrained deep features to detect unconstrained face images. Although the detection performances were improved in [22] and [23], they performed feature extraction, independent of the clustering task. The feature distribution in the extracted feature space cannot be well clustered to detect abnormal data. To conquer the problem, Xie et al. [24] integrated a deep neural network with clustering method, instead of only using the deep network to detect abnormal data. Yang et al. [25] designed an joint objective function including a network reconstruction loss and a clustering loss to perform clustering and network training simultaneously. Similarity, Song et al. [26] took advantage of the coupling nature of clustering and outlier detection to design a deep clustering based fair outlier detection method. However, these deep clustering methods have two main drawbacks. First, the centroid of the clustering layer is randomly initialized before each iteration, resulting in training instability. Second, the threshold of anomaly score is manually set, making it unable to adapt to the change of data distribution.

Motivated by the above analysis, the article proposes a self-supervised memory enhanced deep clustering method (SMEL) to detect abnormal data of WWTP. The proposed SMEL combines the advantages of clustering methods and deep learning methods. Compared with the traditional clustering-based methods, the proposed SMEL does not require the assumption of the shape of the clusters and can capture the nonlinear characteristic of WWTP data through nonlinear dimension reduction. Compared with the deep learning-based methods, the proposed SMEL is a self-supervised method, which does not require a great quantity of labeled data in the training process. The main contributions are summarized as follows.

- 1) A self-supervised deep clustering network is proposed for efficiently capturing nonlinear representations and automatically detecting abnormal data without using any labels.
- 2) An adaptive weight objective function is developed to obtain a robust clustering structure by adopting the adaptive weight mechanism.
- 3) Double memory enhanced modules are designed to enhance training stability and detection accuracy by addressing the problems of initial centroid randomization and abnormal score threshold presetting.

The rest of this article is organized as follows. The problem formulation is presented in Section II. The proposed SMEL is designed in details in Section III. The proposed SMEL is tested on two benchmark datasets and a real WWTP dataset to manifest its performances in Section IV. Finally, Section V concludes this article.

II. PROBLEM FORMULATION

Different from typical supervised detection problem, abnormal data detection for WWTP faces two challenges. One is that obtaining data with anomaly labels is difficult and time-consuming, resulting in the existing supervised learning methods infeasible. The other is that WWTP data are multivariate

TABLE I
LIST OF ACRONYMS

Acronym	Definition
WWTP	Wastewater treatment process
SMEL	Self-supervised memory enhanced deep clustering
SAE	Stacked autoencoders
BSM1	Benchmark simulation model no. 1
CM	Centroid memory module
SM	Score memory module
ST	Storm dataset from BSM1
RA	Rain dataset from BSM1
QH	Dataset collected from Qinghe treatment process
COD	Chemical oxygen demand
NH ₄ N	Ammonia nitrogen
DO	Dissolved oxygen
NO ₃ N	Nitrate nitrogen
PH	Potential of hydrogen
ROC	Receiver operating characteristic
AUC	Area under curve
Acc	Accuracy
F ₁	F-measure
BA	Balanced accuracy
FM	Fowlkes-Mallows
CBLOF	Clustering-based on local outlier factor
COR	Clustering with outlier removal
iForest	Isolation forest
LODA	Lightweight online detection of anomalies
OCSVM	One-class support vector machine
AE	Autoencoder
DEC	Deep embedded clustering
IDEC	Improved deep embedded clustering

time series with strong nonlinearity and time-varying characteristic. Thus, this article focuses on constructing a deep neural network to extract data characteristics without using labels. The network can robustly capture the normal pattern and effectively detect abnormal data of WWTP. The acronyms used in the article are given in Table I.

Consider a matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]^T \in \mathbb{R}^{T \times N}$ for a multivariate time series that collected from N sensors with T time steps. Each instance $\mathbf{x}_i \in \mathbb{R}^N$ is a N -dimensional vector. To detect whether the instance $\mathbf{x}_i$ is abnormal, the data $\mathbf{X}$ are utilized to robustly capture the normal pattern and mark the labels.

Thus, the global objective function of detection task becomes

$$\arg \min_{\mathbf{I}, \boldsymbol{\theta}} F(\mathbf{I}, \boldsymbol{\theta} | \mathbf{X}) \quad (1)$$

where $F(\cdot)$ is the objective function, $\mathbf{I} = [I_1, I_2, \dots, I_T]$ denotes the clustering ids for all instances, and $\boldsymbol{\theta}$ is the network parameters for representations. In such case, if we hold one in $\{\mathbf{I}, \boldsymbol{\theta}\}$ to be fixed, the objective function is decomposed into two alternating steps as

$$\arg \min_{\mathbf{I}} F(\mathbf{I} | \boldsymbol{\theta}, \mathbf{X}) \quad (2)$$

$$\arg \min_{\boldsymbol{\theta}} F(\boldsymbol{\theta} | \mathbf{I}, \mathbf{X}). \quad (3)$$

From the above formulations, it is shown that (2) is a conventional clustering problem based on fixed parameters $\boldsymbol{\theta}$, and (3) is a supervised feature representation learning process based on the cluster labels $\mathbf{I}$.

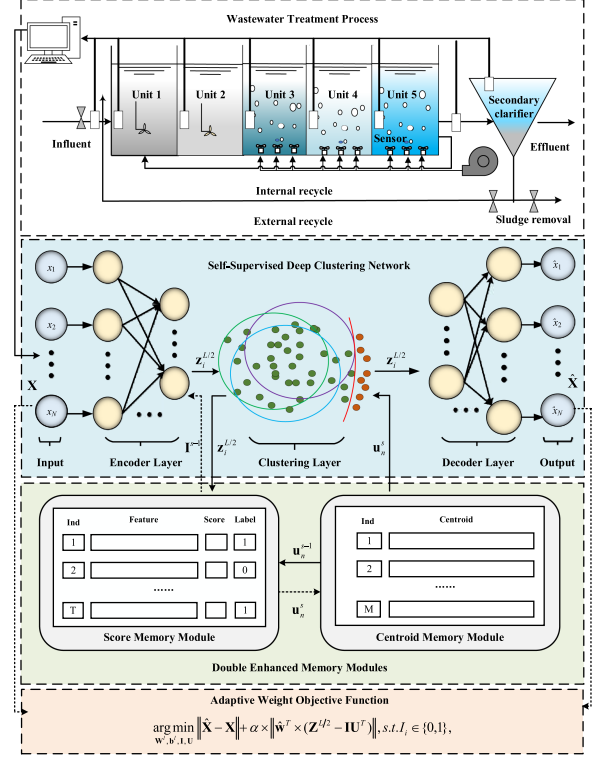


Fig. 1. Overall architecture of the proposed SMEL. (Top) original data $\mathbf{X}$ are collected from WWTP through various sensors. (Middle) The original data $\mathbf{X}$ are fed to the self-supervised deep clustering network to obtain feature points $\mathbf{Z}^{L/2}$. The score memory module provides pseudo-labels $\mathbf{I}^{s-1}$ for the encoder layer and the decoder layer to perform back-propagation and then updates feature points and new pseudo-labels $\mathbf{I}^s$. The centroid memory module is updated by storing the involved centroids $\mathbf{u}_n^s$. (Bottom) the adaptive weight objective function is utilized to optimize the self-supervised deep clustering network.

Once the detection model is determined, the anomaly score $e(\mathbf{x}_i)$ of the instance $\mathbf{x}_i$ can be computed as

$$e(\mathbf{x}_i) = \|F(\mathbf{I} | \boldsymbol{\theta}, \mathbf{x}_i) - \mathbf{u}_n\|^2 \quad (4)$$

where $\mathbf{u}_n$ is the centroid of normal cluster. The higher $e(\mathbf{x}_i)$ is, the more likely it is that the instance $\mathbf{x}_i$ is an abnormal data. To automatically identify abnormal data, the threshold ε is given adaptively, as shown in Section III-D. When the anomaly score $e(\mathbf{x}_i)$ exceeds the given threshold ε , the instance $\mathbf{x}_i$ is identified as abnormal and labeled I_i as 1, otherwise labeled I_i as 0.

III. METHOD

A. Architecture of the Proposed SMEL

The overall architecture of the proposed SMEL is presented in Fig. 1 and is composed of three parts: a self-supervised deep clustering network; an adaptive weight objective function; and double enhanced memory modules.

Specifically, original data $\mathbf{X}$ with the shape of $\mathbb{R}^{T \times N}$ are collected through N sensors installed in WWTP and are fed into the deep clustering network along the temporal dimension. The self-supervised deep clustering network includes an encoder

layer, a clustering layer, and a decoder layer. The instance $\mathbf{x}_i$ is fed into the encoder layer to extract the nonlinear feature representation. The clustering layer captures the robust normal pattern through several iterations. The decoder layer is applied to reconstruct input instance. To optimize the parameters of the self-supervised deep clustering network, the adaptive weight objective function is designed by integrating the reconstruction error and the clustering error. The function is utilized to realize the constraint of feature learning making the clustering structure more suitable for anomaly detection. Meanwhile, to enhance training stability and detection accuracy, double memory enhanced modules, consisting of a centroid memory and a score memory, are proposed through building up recurrent connections with the clustering layer and caching the cluster centroids and feature representations, respectively. The centroid memory module generates a centroid before each iteration by calculating historical clustering centroid. The score memory module is applied to calculate anomaly score thresholds and assign anomaly labels.

B. Self-Supervised Deep Clustering Network

The self-supervised deep clustering network consists of three components: an encoder layer (for dimension reduction and feature extraction), a clustering layer (for feature clustering and abnormal data detection), and a decoder layer (for data reconstruction). The collected WWTP data $\mathbf{X} \in \mathbb{R}^{T \times N}$ are fed into the self-supervised deep clustering network composed of $L + 1$ layers that perform L nonlinear projections, where L is an even number.

1) Encoder Layer: The first $(L/2)$ hidden layers are encoders, which are utilized for compressing each instance $\mathbf{x}_i$ into the latent space. For encoders, the output feature of the l th layer is computed by

$$\mathbf{z}_i^l = f(\mathbf{W}^l \mathbf{x}_i + \mathbf{b}^l) \in \mathbb{R}^{N^l} \quad (5)$$

where $l = 1, 2, \dots, (L/2)$ indexes the layers of encoders, $\mathbf{W}^l$ and $\mathbf{b}^l$ are the weight matrix from the $(l-1)$ th layer to the l th layer and the bias vector of the l th layer, respectively, and N^l is the latent space dimension of l th layer. $f(\cdot)$ is a nonlinear activation function. Thus, for T given instances, the output matrix of the encoder layer $\mathbf{Z}^{L/2} = [\mathbf{z}_1^{L/2}, \mathbf{z}_2^{L/2}, \dots, \mathbf{z}_T^{L/2}]^T \in \mathbb{R}^{T \times L/2}$ is the desired feature representation of original data $\mathbf{X}$. The matrix $\mathbf{Z}^{L/2}$ is fed to the clustering layer for capturing the normal pattern.

2) Clustering Layer: The $(L/2)$ layer serves as the latent feature space and is shared by the encoder layer and the decoder layer. Taking the real WWTP dataset without any labels into consideration, the clustering layer needs to obtain the normal pattern in a self-supervised way. Therefore, the density peak clustering algorithm is selected [27]. The core idea in the density peak clustering algorithm is to calculate the local density ρ_i of each feature points $\mathbf{z}_i^{L/2}$ and the delta distance δ_i of high-density feature points.

The local density ρ_i of each feature point $\mathbf{z}_i^{L/2}$ is defined as

$$\rho_i = \sum_{j=1}^T \chi(d_{ij} - d_c), \quad (6)$$

where $\chi(\cdot)$ is an indicator function. $\chi(x) = 1$, if $x < 0$; otherwise, $\chi(x) = 0$. d_c is a cutoff distance and d_{ij} is the Euclidean distance between each two feature points as

$$d_{ij} = d(\mathbf{z}_i^{L/2}, \mathbf{z}_j^{L/2}) = \|\mathbf{z}_i^{L/2} - \mathbf{z}_j^{L/2}\|. \quad (7)$$

The delta distance δ_i between the feature point $\mathbf{z}_i^{L/2}$ and any other feature points with higher density is defined as

$$\delta_i = \begin{cases} \min_{j: \rho_j > \rho_i} (d_{ij}), & \text{if } \exists j \text{ s.t. } \rho_j > \rho_i \\ \max_j (d_{ij}), & \text{otherwise} \end{cases}. \quad (8)$$

When the local density ρ_i and the delta distance δ_i for each feature point have been calculated, the feature points with a high ρ_i and high δ_i are considered as cluster centers. In real WWTP dataset, the number of normal data is much larger than the number of abnormal data. Thus, the proposed SMEL selects the centroid $\mathbf{u}_n \in \mathbb{R}^{L/2 \times 1}$ with the highest density as the centroid of normal cluster and the centroid $\mathbf{u}_b \in \mathbb{R}^{L/2 \times 1}$ with the lowest density as the centroid of abnormal cluster. After finding the cluster centers, each remaining feature point is assigned to the same cluster as its nearest neighbor of higher density. We mark the ids of the normal clustering and the abnormal clustering to 0 and 1, respectively.

3) Decoder Layer: The last $(L/2)$ layers are decoders, which are applied to reconstruct the instance $\mathbf{x}_i$ from the latent space. For decoders, the output of the l' th layer is compute by

$$\mathbf{z}_i^{l'} = g(\mathbf{W}^{l'} \mathbf{z}_i^{l'-1} + \mathbf{b}^{l'}) \in \mathbb{R}^{N^{l'}} \quad (9)$$

where $l' = (L/2) + 1, (L/2) + 2, \dots, L$ indexes the layers of the decoder, $\mathbf{W}^{l'}$ is the weight matrix from $(l'-1)$ th layer to the l' th layer, and $\mathbf{b}^{l'}$ is the bias of the l' th layer. In this article, the weight $\mathbf{W}$ is equivalent to the transposition of the weight matrix $\mathbf{W}'$. The nonlinear activation function $g(\cdot)$ can be same as $f(\cdot)$ or another different nonlinear function. Thus, the corresponding reconstruction of instance $\mathbf{x}_i$ is computed by

$$\hat{\mathbf{x}}_i = g(\mathbf{W}^{L'} \mathbf{z}_i^{L'} + \mathbf{b}^{L'}) \in \mathbb{R}^N. \quad (10)$$

The output matrix of the decoder layer $\hat{\mathbf{X}} = [\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \dots, \hat{\mathbf{x}}_T]^T \in \mathbb{R}^{T \times N}$ is the corresponding reconstruction of data $\mathbf{X}$.

Remark 1: The self-supervised deep clustering network has powerful capabilities on feature extraction and data representation since encoders and decoders have strong nonlinear mapping characteristic. Meanwhile, the clustering layer captures the normal pattern without requiring any labels, which is conducive to WWTP automatic detection.

C. Adaptive Weight Objective Function

1) Joint Objective Function: Taking the reciprocal relationship between abnormal data and clustering into account, the

degree of abnormality for each instance is associated with the distance to normal cluster centroid. An adaptive weight mechanism is proposed to enhance the effectiveness of abnormal data detection by re-emphasizing the contributions of possible normal data while mitigating negative impacts of abnormal data. Specifically, feature points close to the centroid of normal category are assigned higher weights while those far away from centroids, identified as potential abnormal candidates, are assigned lower weights. Thus, an adaptive weight $\hat{\mathbf{w}}_i$ of the feature point $\mathbf{z}_i^{L/2}$ is calculated as

$$\hat{w}_i = 1 - \frac{\exp(d(\mathbf{z}_i^{L/2}, \mathbf{u}_n))}{\sum_{i=1}^T \exp(d(\mathbf{z}_i^{L/2}, \mathbf{u}_n))}. \quad (11)$$

The vector $\hat{\mathbf{w}} = [\hat{w}_1, \hat{w}_2, \dots, \hat{w}_T] \in \mathbb{R}^{T \times 1}$ denotes the adaptive weights of all feature points $\mathbf{Z}^{L/2}$. The adaptive weight mechanism enables the normal data with high weights to capture the intrinsic cluster structure and guarantees the performance of abnormal data detection.

To extract the suitable clustering structure and robustly detect abnormal data, the joint objective function with the adaptive weight mechanism is proposed to realize the constraint of detection objective on feature extraction. Based on three components of the self-supervised deep clustering network, the objective function is formally stated as

$$\arg \min_{\mathbf{W}^l, \mathbf{b}^l, \mathbf{I}, \mathbf{U}} \underbrace{\|\hat{\mathbf{X}} - \mathbf{X}\|}_{\mathcal{J}_1} + \alpha \underbrace{\|\hat{\mathbf{w}}^T \times (\mathbf{Z}^{L/2} - \mathbf{I}\mathbf{U}^T)\|}_{\mathcal{J}_2}, s.t. I_i \in \{0, 1\} \quad (12)$$

where $\mathbf{U} = [\mathbf{u}_n, \mathbf{u}_b] \in \mathbb{R}^{L/2 \times 2}$ denotes the clustering centroids of which each column represents one cluster centroid, $\mathbf{I} \in \mathbb{R}^{T \times 2}$ denotes the indicator matrix of which each row denotes the binary label, and α is tradeoff parameter between the reconstruction error and the cluster error. $\mathcal{J}_1$ and $\mathcal{J}_2$ are designed for different goals. $\mathcal{J}_1$ is the mean absolute error term to ensure the minimum reconstruction error which avoiding the trap of trivial solutions by nonlinear dimensionality reduction. $\mathcal{J}_2$ is a clustering regularization aiming to discover the clustering-friendly latent space.

2) Objective Function Optimization: The optimization process includes two stages: pretraining stage and fine-tuning stage. The pretraining stage minimizes the reconstruction loss of encoders and decoders by using stochastic gradient descent and backpropagation to obtain feature points of original data $\mathbf{X}$. Since the training of encoders and decoders is not clustering-specific, thus the feature points are not necessarily suitable for abnormal data detection.

In the fine-tuning stage, we alternately optimize (12) with respect $\mathbf{W}^l$, $\mathbf{b}^l$, $\mathbf{I}$, and $\mathbf{U}$ to obtain a robust clustering structure. Since the network parameters have been determined after pre-training, (12) is identical to the objective of clustering algorithm expect for the parameters to optimize. The objective is expressed as

$$\arg \min_{\mathbf{I}, \mathbf{U}} \|\hat{\mathbf{X}} - \mathbf{X}\| + \alpha \|\hat{\mathbf{w}}^T \times (\mathbf{Z}^{L/2} - \mathbf{I}\mathbf{U}^T)\|, s.t. I_i \in \{0, 1\} \quad (13)$$

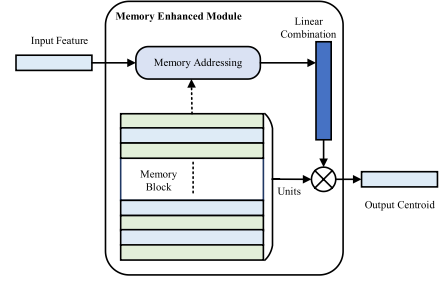


Fig. 2. Architecture of the memory module.

which separates feature points by alternately updating cluster centroids $\mathbf{U}$ and ids $\mathbf{I}$. After the pretraining stage, the density peak clustering is used to initialize the cluster centroid $\mathbf{U}^0 = [\mathbf{u}_n^0, \mathbf{u}_b^0] \in \mathbb{R}^{L/2 \times 2}$. The clustering centroid is stored to a centroid memory module, which aims to provide the centroid $\mathbf{U}^s = [\mathbf{u}_n^s, \mathbf{u}_b^s] \in \mathbb{R}^{L/2 \times 2}$ for the subsequent iteration processes, as shown in Section III-D. $\mathbf{u}_n^s$ is the centroid of the normal cluster at the s th iteration. In each iteration, each feature point is assigned with a pseudolabel I_i^s and is stored into a score memory module, as shown in Section III-D.

When $\mathbf{U}$ and $\mathbf{I}$ are fixed, we read pseudo-labels from the score memory module. Thus, (12) is expressed as

$$\arg \min_{\mathbf{W}^l, \mathbf{b}^l} \|\hat{\mathbf{X}} - \mathbf{X}\| + \alpha \|\hat{\mathbf{w}}^T \times (\mathbf{Z}^{L/2} - \mathbf{I}\mathbf{U}^T)\|, s.t. I_i \in \{0, 1\}. \quad (14)$$

The objective can be interpreted as the objective of supervised learning for the encoders and decoders. Because it is cluster-specific, it can build a robust clustering structure for abnormal data detection.

Remark 2: By optimizing the adaptive weight objective function, the proposed SMEL enhances the sensitivity to the noise and obtains a robust clustering structure because the learning of the clustering layer is more concerned with the normal pattern rather than just blindly minimizing the reconstruction error.

D. Double Memory Enhanced Modules

To enhance training stability and detection accuracy, double memory enhanced modules, consisting of a centroid memory and a score memory, are developed. The centroid memory module stores the cluster centroid of each iteration process, while the score memory module stores feature points and pseudo-labels of the entire dataset. Pseudolabels and network parameters are updated alternately during uninterrupted iterations. A memory enhanced module, including a feature extractor I_f , a memory block Mb , a generalization process G , and an output feature O_f , was first designed in [28]. The generalization process G updates the memory block Mb with the input feature I_f during training process. The output feature O_f is a new feature rebuilt by the memory block Mb . The architecture of the memory module is shown in Fig. 2.

1) Centroid Memory Module: The centroid memory module $CM \in \mathbb{R}^{M \times L/2}$ is designed to record the involved centroids, including those in which new centroids join and those from which old centroids leave. Compared with the random initialization

of centroid, the centroid memory module provides the mean feature of the involved centroids for each iteration. The centroid preserves the historical clustering information and improves the clustering efficiency. Specifically, the centroid memory module with M units builds up a recurrent connection with the clustering layer, stores cluster centroids to memory block in a dynamic way, and produces the centroid of each iteration via a liner combination of memory units. When the number of centroids reaches M , the output feature serves as the centroid of the next iteration as

$$\mathbf{U}^s = \frac{\sum_{m=0}^M \mathbf{U}^{s-M}}{M} \quad (15)$$

where $\mathbf{U}^{s-M}$ is the clustering centroids of the $(s-M)$ th iteration. When the number of centroids does not reach M , the centroid is calculated via the density peak clustering algorithm.

2) Score Memory Module: The score memory module $SM \in \mathbb{R}^{T \times (L/2+2)}$ is designed to provide pseudo-labels for performing back-propagation to update the encoders and decoders. Specifically, the score memory module with T unites builds up recurrent connections with the encoder layer and the centroid memory module, respectively. The module is updated by updating the feature points from the encoder layer and is assigned new pseudo-labels by calculating anomaly score.

In each iteration, the feature points from the encoder layer and the centroid from the centroid memory module are utilized to calculate the anomaly scores and mark the pseudolabels. The anomaly score $e(\mathbf{z}_i^{L/2})$ is calculated as

$$e(\mathbf{z}_i^{L/2}) = \left\| \mathbf{z}_i^{L/2} - \mathbf{u}_n^s \right\|^2 \quad (16)$$

where $\mathbf{u}_n^s$ is the centroid of the normal cluster in the s th iteration. To judge the abnormal degree of feature points, the anomaly score thresholds $\varepsilon_{\max}$ and $\varepsilon_{\min}$ are calculated as

$$\varepsilon_{\max} = \mu + 3\sigma \quad (17)$$

$$\varepsilon_{\min} = \mu - 3\sigma \quad (18)$$

where μ and σ are the mean and standard deviation of abnormal scores, respectively, which are calculated as

$$\mu = \frac{1}{T} \sum_{i=1}^T e(\mathbf{z}_i^{L/2}) \quad (19)$$

$$\sigma = \sqrt{\frac{1}{T} \sum_{i=1}^T \left(e(\mathbf{z}_i^{L/2}) - \mu \right)^2} \quad (20)$$

When the anomaly score $e(\mathbf{z}_i^{L/2})$ exceeds thresholds $\varepsilon_{\min}$ or $\varepsilon_{\max}$, the feature point $\mathbf{z}_i^{L/2}$ is judged to be abnormal and marks its anomaly label I_i^s as 1. Otherwise, the anomaly label I_i^s is marked as 0. The criteria are as

$$I_i^s = \begin{cases} 1, & \text{if } e(\mathbf{z}_i^{L/2}) \leq \varepsilon_{\min} \\ 0, & \text{if } \varepsilon_{\min} < e(\mathbf{z}_i^{L/2}) < \varepsilon_{\max} \\ 1, & \text{if } e(\mathbf{z}_i^{L/2}) \geq \varepsilon_{\max} \end{cases} \quad (21)$$

Remark 3: The centroid memory module stores the clustering centroid to solve the problem of initial centroid randomization.

Algorithm 1: Self-Supervised Memory Enhanced Deep Clustering.

Input: Original data $\mathbf{X}$, Hyper-parameter α , Maximum iterations S , Centroid memory module units M

Output: The network parameters θ , The centroid $\mathbf{u}_n^s$

- 1: Pretrain the parameters of stacked autoencoders for $\mathbf{W}^l, \mathbf{W}'^l, \mathbf{b}^l, \mathbf{b}'^l$
 - 2: Initialize $\mathbf{U}^0, \mathbf{I}^0$ by deploying density peak clustering on $f(\mathbf{x}_i)$, i.e., the output the pretrained encoder
 - 3: Store the centroid $\mathbf{U}^0$ to the centroid memory module CM
 - 4: **for** $s = 1$ to S **do**
 - 5: Calculate the weight $\hat{\mathbf{w}}_i$ of each feature point $\mathbf{z}_i^{L/2}$ via (11)
 - 6: Update parameters $\mathbf{W}^l, \mathbf{W}'^l, \mathbf{b}^l, \mathbf{b}'^l$ by optimizing (14) with stochastic gradient descent
 - 7: **if** $\text{len}(CM) < M$
 - 8: Update $\mathbf{U}^s$ by deploying density peak clustering
 - 9: **else**
 - 10: Update $\mathbf{U}^s$ via (15)
 - 11: **end if**
 - 12: Store centroid $\mathbf{U}^s$ to the centroid memory module
 - 13: Update pseudo-labels $\mathbf{I}^s$ via (16)–(21)
 - 14: Store pseudo-labels to the score memory module SM
 - 15: **if** convergence condition is met **then**
 - 16: Stop training
 - 17: **end if**
 - 18: **end for**
 - 19: **return** The network parameters θ , The centroid $\mathbf{u}_n^s$
-

The score memory module stores feature points and corresponding pseudolabels to provide supervision information for performing back-propagation of encoders and decoders. The pseudolabels are marked by comparing each anomaly score and the adaptive thresholds. So far, the detailed procedure of the proposed SMEL can be summarized as Algorithm 1.

IV. EXPERIMENTS

A. Experimental Settings

1) Datasets: To verify the effectiveness of the proposed SMEL, two commonly-used datasets and a real dataset are selected. The detailed information of three datasets is given in Table II. Note that abnormal data are a small part of a dataset, which is consistent with the actual phenomenon.

a) ST: The storm dataset contained 1335 instances and 10 attributes at a 15-min interval. The dataset was obtained from the benchmark simulation model no.1 (BSM1). BSM1 was proposed by the international water association task group and is considered as a recognized platform in field of wastewater treatment.

b) RA: The rain dataset consists of 10 attributes and 1335 instances at a 15-min interval and is selected from BSM1.

TABLE II
CHARACTERISTICS OF DATASETS

Statistics	BSM1		Real
	ST	RA	QH
Instance	1335	1335	8000
Attribute	10	10	28
Train rate	40%	40%	20%
Anomaly rate	5.83%	17.2%	22.6%
Anomaly period	[77.25, 79.25] [127.75, 135.75] [139.75, 140.75]	[67, 84] [90.5, 107]	[3000, 4400] [3], [10]
Anomaly attribute	[10]	[10]	[1], [15], [16], [18], [19], [20]

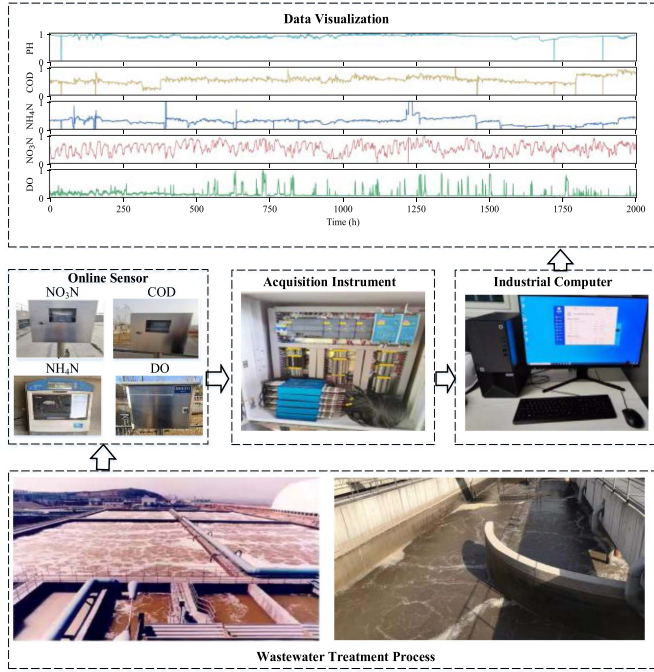


Fig. 3. Schematic diagram of Qinghe wastewater treatment plant.

c) QH: As a large-scale dataset, QH dataset was collected from a wastewater treatment plant located in Qinghe Beijing, China over a 25-day period every 15 mins. The dataset consists of 28 attributes and 8000 instances. As shown in Fig. 3, collection of the QH dataset is implemented through deploying sensors on all the key process units, such as chemical oxygen demand (COD), ammonia nitrogen (NH_4N), dissolved oxygen (DO), nitrate nitrogen (NO_3N), potential of hydrogen (PH), and so on. Then, the acquisition instrument is utilized to transmit data to the industrial computer. Finally, the industrial computer provides data visualization and implements actuator control decisions if necessary. The complete dataset is stored to a secured database that can be accessed by the industrial partners and the research team through the industrial computer. The data visualization in Fig. 3 shows that NO_3N time series has periodic trend to some extent. For DO time series, the data are relatively stable during the first 500 hours and then fluctuated sharply. PH, COD, and NH_4N obviously contain abnormal values presenting random characteristic.

These data reflect the operation status of the actual WWTP, such as sensor faults, communication losses, and changed environment.

2) Baseline Methods: Clustering-based methods including clustering-based on local outlier factor (CBLOF) [29] and clustering with outlier removal (COR) [30], model-based methods including isolation forest (iForest) [31], lightweight online detection of anomalies (LODA) [32], one-class support vector machine (OCSVM) [33], and autoencoder (AE) [34], and deep clustering methods including deep embedded clustering (DEC) [24] and improved deep embedded clustering (IDEC) [35] are chosen as baseline methods. All the codes are implemented by Python 3.9. In practice, the parameters of these anomaly detection methods are set as follows: for iForest, the subsampling size and the tree number are 256 and 100, respectively; the number of clusters for CBLOF and COR are set to be 2; for LODA, the number of bins and random cuts are 10 and 100, respectively; the radial basis function of OCSVM is set to be 0.5; for AE, DEC, and IDEC, the hidden neuron structures are 16-8-8-16, the maximum iteration epoch are set to be 250, and the learning rate is 0.1; for the proposed SMEL, three fully-connected layers are set and the network structure for training the QH dataset is set to be 12-9-6-9-12 and the network structures for training the ST and RA datasets are set to be 8-6-4-6-8. The maximum iteration epoch is set to be 250. The learning rate of the cluster layer is set to be 0.001 and the learning rates of the encoders and decoders are set as 0.1. During evaluation process, the mean and standard deviation for all methods by repeating their experiments on 20 random seeds.

3) Evaluation Criteria: For quantitatively evaluating the detection performance, accuracy (Acc), F-measure (F_1), balanced accuracy (BA), Fowlkes–Mallows (FM), receiver operating characteristic (ROC), and area under curve (AUC) are selected. The definitions of five metrics are list.

a) Acc: The metric is the proportion of the correct prediction in all prediction samples as

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (22)$$

where TP, TN, FP, and FN are true positive, true negative, false positive, and false negative, respectively.

b) F_1 : It is a harmonic mean of precision and recall as

$$F_1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \quad (23)$$

c) BA: It indicates the average of recall and the equivalent of recall calculated with respect to the category of non-anomalous values as

$$\text{BA} = \frac{1}{2} \times \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right) \quad (24)$$

d) FM: It is defined as the geometric mean of precision and recall. FM is defined by

$$\text{FM} = \frac{\text{TP}}{\sqrt{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN})}} \quad (25)$$

e) ROC: ROC curve is a graph consisting of the false positive rate (FPR) on the x -axis and the true positive rate

TABLE III
EXPERIMENTAL RESULTS OF ANOMALY DETECTION ON THREE DATASETS FOR ALL METHODS

Dataset	Metric	Clustering-based		Model-based				Deep clustering		Ours
		CBLOF [29]	COR [30]	iForest [31]	LODA [32]	OCSVM [33]	AE [34]	DEC [24]	IDEC [35]	SMEL
ST	Acc	0.86 ± 0.01	0.94	0.88 ± 0.01	0.87 ± 0.01	0.90	0.89 ± 0.01	0.93 ± 0.02	0.92 ± 0.03	0.95 ± 0.01
	F_1	0.39 ± 0.02	0.43	0.37 ± 0.09	0.05 ± 0.02	0.42	0.42 ± 0.01	0.59 ± 0.20	0.50 ± 0.17	0.61 ± 0.08
	BA	0.89 ± 0.01	0.71	0.61 ± 0.08	0.49 ± 0.01	0.84	0.83 ± 0.01	0.83 ± 0.10	0.88 ± 0.11	0.89 ± 0.04
	FM	0.85 ± 0.01	0.94	0.88 ± 0.01	0.88 ± 0.01	0.89	0.89 ± 0.01	0.93 ± 0.02	0.92 ± 0.03	0.95 ± 0.01
RA	Acc	0.73 ± 0.03	0.73	0.83 ± 0.01	0.77 ± 0.01	0.78	0.81 ± 0.03	0.80 ± 0.05	0.79 ± 0.05	0.83 ± 0.05
	F_1	0.24 ± 0.01	0.21	0.48 ± 0.02	0.15 ± 0.05	0.37	0.55 ± 0.12	0.21 ± 0.26	0.36 ± 0.20	0.60 ± 0.12
	BA	0.54 ± 0.01	0.52	0.62 ± 0.01	0.47 ± 0.01	0.63	0.63 ± 0.08	0.52 ± 0.12	0.58 ± 0.10	0.65 ± 0.07
	FM	0.72 ± 0.01	0.73	0.81 ± 0.01	0.78 ± 0.01	0.74	0.75 ± 0.01	0.73 ± 0.04	0.78 ± 0.04	0.81 ± 0.05
QH	Acc	0.32 ± 0.01	0.69	0.32 ± 0.01	0.39 ± 0.09	0.33	0.30 ± 0.01	0.54 ± 0.11	0.57 ± 0.08	0.70 ± 0.04
	F_1	0.44 ± 0.01	0.14	0.45 ± 0.01	0.44 ± 0.01	0.42	0.44 ± 0.01	0.44 ± 0.01	0.42 ± 0.05	0.47 ± 0.05
	BA	0.51 ± 0.01	0.51	0.52 ± 0.01	0.52 ± 0.04	0.48	0.50 ± 0.01	0.50 ± 0.01	0.56 ± 0.05	0.60 ± 0.03
	FM	0.72 ± 0.01	0.72	0.73 ± 0.01	0.65 ± 0.08	0.67	0.74 ± 0.01	0.73 ± 0.01	0.57 ± 0.03	0.79 ± 0.04

The bold values show the best results among all methods.

(TPR) on the y-axis, where $TPR = TP/(TP + FN)$ and $FPR = TP/(FP + TN)$. FPR and TPR are calculated and plotted at various threshold settings of the confidence score of the detection outputs.

f) AUC: It reflects the performance of the classification model. The closer AUC value is to 1, the better the classification is.

B. Experimental Results

1) Comparison of Detection Results: The results of the ST dataset are shown in the first section of Table III. Generally, the neural network based methods outperform the nonneural network based methods in most metrics. The relatively low F_1 for iForest and LODA come with a price with high Acc, which means that many instances that do not contain outliers are mistakenly considered as outliers. On average, the proposed SMEL achieves the best results for Acc, F_1 , BA, and FM, showing good detection performance overall in comparison to eight baseline methods. For the RA dataset, it can be noticed in the middle of Table III. The proposed SMEL achieves the highest accuracy on all metrics, demonstrating the model functionality. Moreover, the efficacy of the proposed SMEL is satisfactory as well with the corresponding FM rate arriving at up to 81%.

2) Visualization of Detection Results: To further investigate the detection performance of the proposed SMEL in detail, the anomaly scores on the best five methods under the ST dataset are shown in Fig. 4. For the proposed SMEL, the mean value and standard deviation of the anomaly score are 0.117 and 0.156, respectively. Therefore, anomaly score thresholds $\varepsilon_{\max}$ and $\varepsilon_{\min}$ are calculated as 0.585 and -0.351 , respectively. Moreover, the original values shown in Fig. 4(a) are all positive. Due to the space limitation, only one positive cutting threshold is shown in the article.

The anomaly scores of OCSVM, AE, and DEC in Fig. 4 are not stable and contain many false negatives, which indicates that

TABLE IV
ANALYSIS OF DIFFERENT MODULES ON THREE DATASETS

Dataset	Module	Acc	F_1	BA	FM
ST	No weight	0.89	0.26	0.57	0.79
	No memory	0.91	0.35	0.66	0.90
	SMEL	0.94	0.55	0.82	0.95
RA	No weight	0.59	0.37	0.63	0.62
	No memory	0.60	0.42	0.70	0.62
	SMEL	0.83	0.60	0.65	0.81
QH	No weight	0.48	0.32	0.48	0.60
	No memory	0.38	0.37	0.46	0.61
	SMEL	0.70	0.47	0.60	0.63

The bold values show the best results.

OCSVM, AE, and DEC have poor anomaly detection performances. Compared with Fig. 4(e) and (f), the curve of anomaly score produced by the proposed SMEL is similar to the original time series. The result demonstrates that the proposed SMEL shows relatively superior detection performance than IDEC. Furthermore, in the shaded regions, the proposed SMEL is well aligned with the original anomaly dimension. The main reason is that the adaptive weight mechanism assigns lower weights to these feature points far away from normal centroid. After several iterations, the anomalous scores of abnormal data are higher than the threshold $\varepsilon_{\max}$.

3) Analysis of Different Modules in SMEL: To evaluate our design choices, the effect of each module in the proposed SMEL is studied by removing model components. Specifically, we remove the adaptive weight mechanism (no weight), and double memory modules (no memory), respectively.

In the first ablation experiment, the proposed SMEL has larger Acc, F_1 , BA, and FM meaning it has better detection performance, as given in Table IV. The adaptive weight mechanism effectively eliminates impact of abnormal data on the extraction of normal pattern.

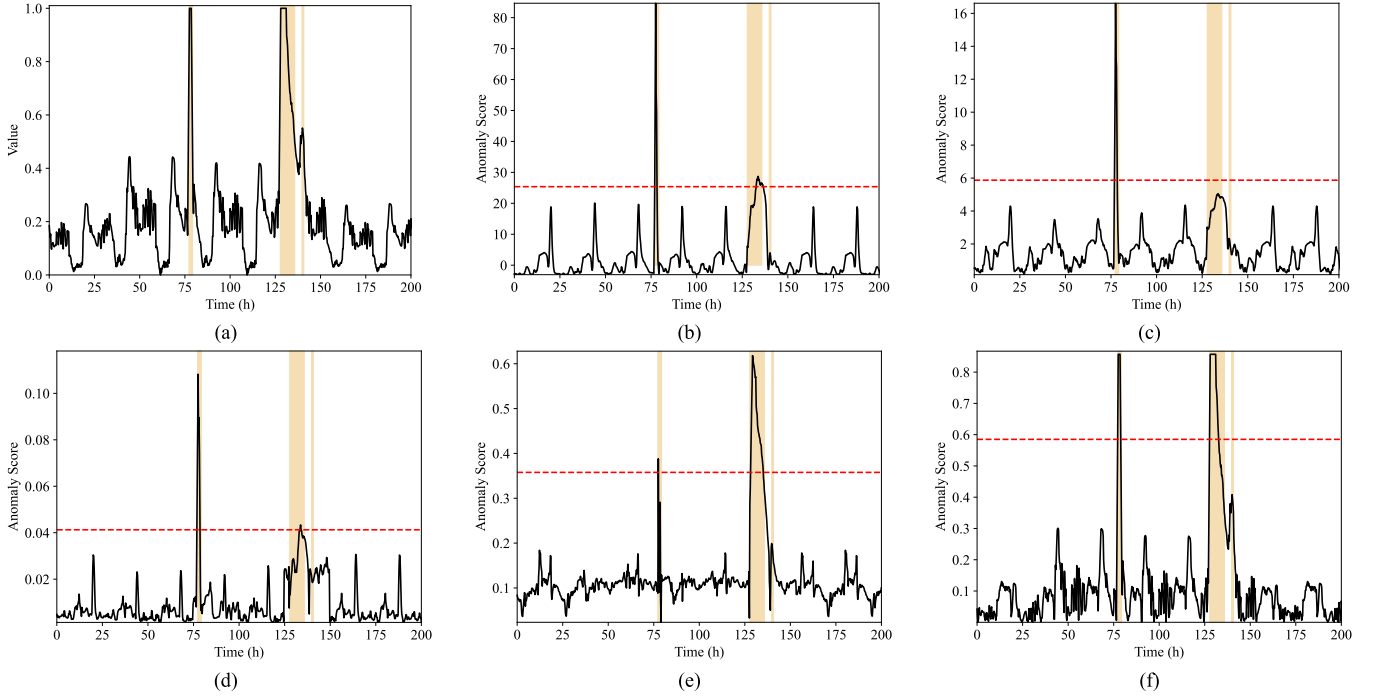


Fig. 4. Visualization of anomaly scores on the ST dataset on the best five methods. The shaded regions represent anomaly periods. The red dash line is the cutting threshold of abnormal data. (a) the anomaly dimension. (b) One-class support vector machine. (c) Autoencoders. (d) Deep embedded clustering. (e) Improved deep embedded clustering. (f) Self-supervised memory enhanced deep clustering method.

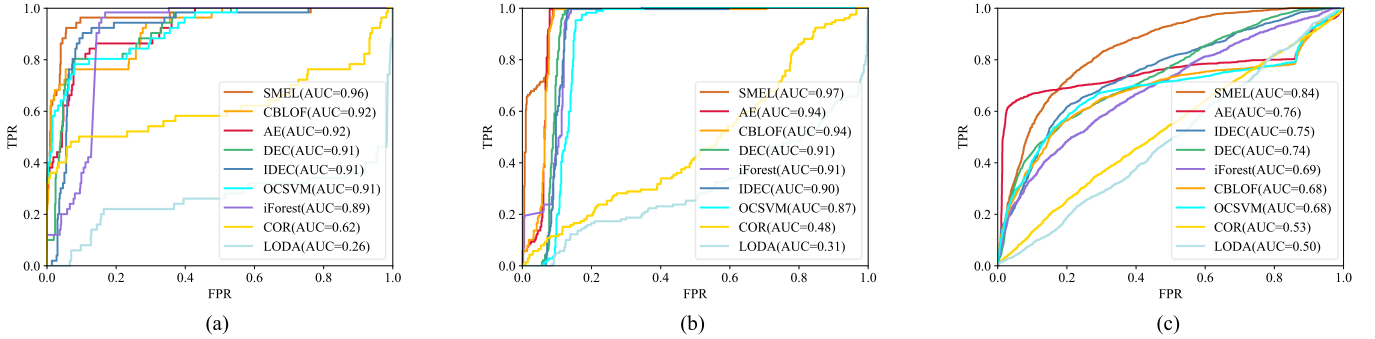


Fig. 5. ROC curves on three WWTP datasets. (a) ST. (b) RA. (c) QH.

In the second ablation experiment, the model with all components achieves the best results in for every modification on almost all metrics. The only exception is BA metric for the RA dataset. Acc values of the proposed SMEL under three WWTP datasets are higher, which means the proposed SMEL has good classification performance. In sum, double memory modules help the proposed SMEL to learn more useful features and improve the performance of anomaly detection.

4) Evaluation on Self-Supervised Deep Clustering Network: To evaluate the performance of the self-supervised deep-clustering network, the standard ROC curve and AUC are used. By changing the threshold, ROC curves and AUC values on three WWTP datasets are obtained, as shown in Fig. 5.

The proposed SMEL clearly outperforms the other eight methods. For the ST and RA datasets, AUC values of AE, IDEC, and DEC are all above 0.9. For QH dataset, AUC values of AE, IDEC, and DEC are all above 0.7. While AUC values of COR

and LODA are around [0.3, 0.65] on three datasets. Furthermore, the ROC curves of the proposed SMEL under three WWTP datasets show that TPR values quickly reach 0.9, 0.9, and 0.8, respectively, during the process of continuously reducing the threshold. These results show that the self-supervised deep clustering network is more robust than other comparison methods.

5) Evaluation on the Clustering Performance: To validate the effectiveness of the clustering layer, the t -distributed stochastic neighbor embedding (t -SNE) is utilized to visualize the latent space in different iterations. Due to the page limitation, we only show the feature distribution under the ST dataset in Fig. 6. The black points denote the normal feature points and the blue points denote the abnormal feature points.

The distribution of feature points in Fig. 6(a) is relatively discrete and all feature points are marked abnormal, implying the difficulty of the anomaly detection. As the training process goes, say at the 70th epoch, most feature points have crowded

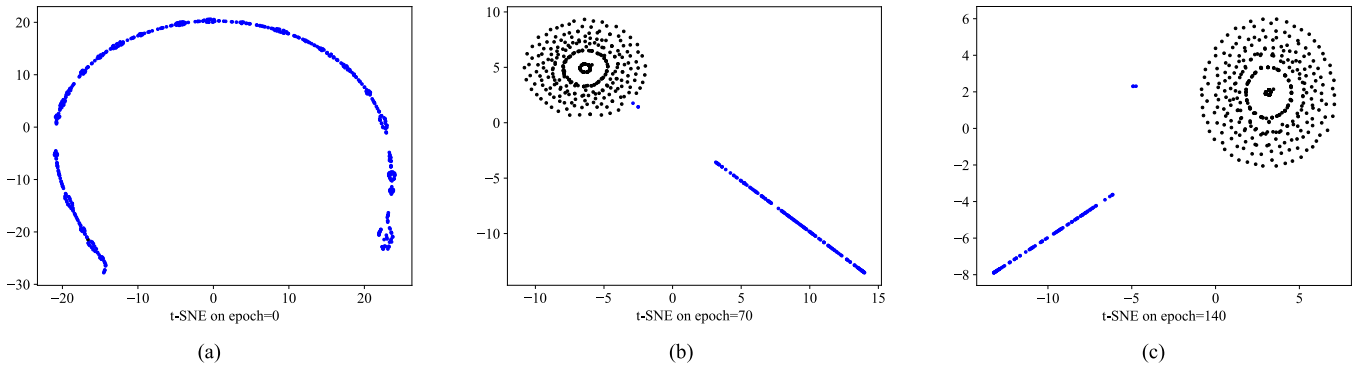


Fig. 6. t-SNE visualization of embedded features on ST dataset. The black points denote the normal feature points and the blue points denote the abnormal feature points.

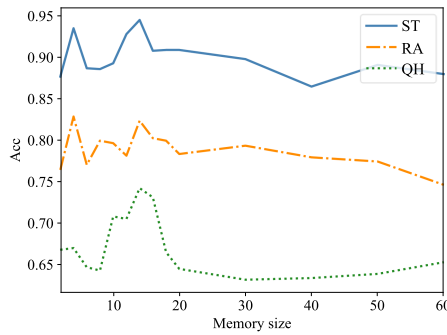


Fig. 7. Robustness of the memory size.

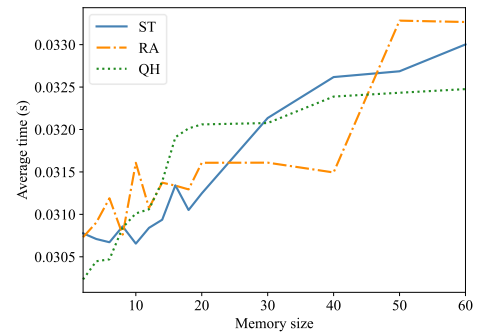


Fig. 8. Average running time on three WWTP datasets.

together and formed the normal cluster. At the 140th epoch, feature points reach stability, which clearly distinguishes between normal features and abnormal features and obtains the centroid of the normal cluster. The phenomenon indicates that the clustering layer can obtain the normal pattern by iteratively updating cluster centroids.

6) Parameter Sensitivity Analysis: Finally, we also conduct experiments on three WWTP datasets to investigate the sensitivity of the memory size M and the hyperparameter α .

Acc values of the proposed SMEL with different memory size on three WWTP datasets are shown in Fig. 7. Obviously, Acc values with the memory module are higher than those without memory modules. It proves again the effectiveness of the memory module. Moreover, the proposed SMEL has the best detection performance when the memory size M is set to be 14. And with an increasing the memory size M , the performance of the proposed SMEL shows a relatively stable trend. Given a suitable memory size M , the proposed SMEL can robustly produce plausible results.

The average running time for detecting each instance is presented in Fig. 8. The proposed SMEL averagely takes 0.032 seconds for abnormal detection of one instance. In addition, with an increasing the memory size M , the average running time varies by about 0.005s, which induces little additional computation time. Compared to remove the memory module, the proposed SMEL boosts the overall performances by a large margin while introducing acceptable running time. In addition, the running time is much less than the specific interval, normally 15 min.

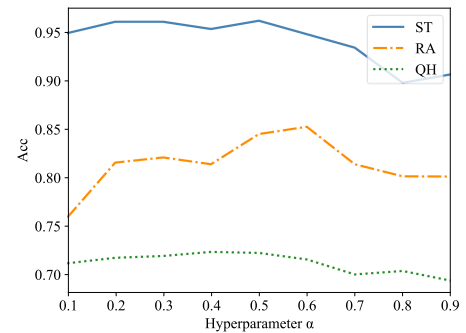


Fig. 9. Hyperparameter α on three WWTP datasets.

Thus, the proposed SMEL can be extended to online detection for WWTP.

The detection performances with different hyperparameter α under three WWTP dataset are shown in Fig. 9. For three WWTP datasets, when $\alpha = 0$, Acc equals 0.948, 0.759, and 0.712, respectively. With the increase of α , Acc gradually increases. When $\alpha = 0.5$, the proposed SMEL obtains the highest Acc under the ST and QH datasets. And when $\alpha = 0.6$, the proposed SMEL obtains the highest Acc under the RA dataset. Thereafter, when α increases, Acc gradually decreases. These results indicate that the parameter α plays a balance role between reconstruction error and clustering error. When α is set to be a large value, the proposed SMEL will have an unsatisfactory performance due to poor reconstruction.

V. CONCLUSION

In this article, a self-supervised memory enhanced deep clustering method (SMEL), including a self-supervised deep clustering network, an adaptive weight objective function, and double memory enhanced modules, was proposed for detecting abnormal data in WWTP without using any labels. The self-supervised deep clustering network with the adaptive weight objective function projected time series into a latent subspace to extract nonlinear structure and iteratively updated the clustering layer to obtain the normal pattern by taking into the negative impact of noise account. Furthermore, the double memory enhanced modules were used to produce the centroids before each iteration for enhancing training stability and provide pseudo-labels for updating the parameters of encoders and decoders. Experimental results demonstrated that the proposed SMEL achieved superior anomaly detection accuracy over eight competitors under three WWTP datasets, and obtained an acceptable running time that was much less than the specific interval, facilitating the promotion of online abnormal detection scenarios. In the future, we plan to extend our work in two directions. First, we will extend the deep clustering algorithm to online scenarios for WWTP. Second, an optimization strategy will be designed to improve detection efficiency.

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